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Trust or reach? Word-of-mouth diffusion under perceived risk: An agent-based simulation

¿Confianza o alcance? Difusión boca a boca bajo riesgo percibido: una simulación basada en agentes

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Abstract

Objective: To examine how prioritizing strong ties, weak ties, or mixed contact rules affects word-of-mouth diffusion under different levels of perceived risk, and to determine how much of the result depends on the assumed link between tie strength and conversion.

Design/Methodology/Approach: A theoretical Python agent-based model represents 600 consumers in six communities over 60 periods. Five strategies and three risk levels produce 1,500 base runs in 100 matched blocks. Additional experiments (i) decouple or invert the association between tie strength and trust, cooperation and commitment while preserving their marginal distributions, (ii) impose a fixed global budget of referral attempts, and (iii) vary community mixing, seeding pattern and a continuous strong/weak portfolio weight. Paired bootstrap intervals quantify Monte Carlo uncertainty. No parameters are calibrated with real data.

Results/Discussion: When relational attributes covary with tie strength, strong-tie priority generates the most new adopters at every risk level (149.58, 115.70 and 89.33, versus 23.70, 18.50 and 14.26 for weak-tie/bridge priority) and keeps its advantage under equal realized effort. This advantage is not an emergent property of the network: when attributes are independent of tie strength, strong- and weak-tie priority are statistically indistinguishable in adoption, conversion and coverage, while weak-tie priority reaches more previously unseeded communities; when the association is inverted, weak-tie priority dominates. The advantage of strong ties becomes resolvable at a rank-coupling weight of roughly 0.1–0.2. No intermediate mixture outperforms pure strong-tie priority.

Conclusions: The ranking of contact rules is governed by how strongly tie strength is assumed to determine transmission, trust and commitment, and not by tie strength or bridge access per se. The observed ranking applies to the specified model and network conditions.

Originality/Value: The study separates adoption volume, individual coverage, community penetration and conversion; separates advantages encoded in behavioral assumptions from emergent ones; and locates the coupling level at which the strong-tie advantage appears.

Keywords: word of mouth, agent-based modeling, tie strength, perceived risk, entrepreneurial marketing, social networks

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1. Introduction

Resource-constrained ventures may depend on existing relationships to initiate word-of-mouth diffusion. A practical allocation problem follows: should limited referral effort concentrate on close contacts, seek contacts outside established circles, or combine both? Entrepreneurial marketing offers a useful setting because it connects opportunity seeking, customer relationships and the creative use of limited resources (Morris et al., 2002). The model addresses referral effort, not firm performance or financial returns.

The apparent choice between trust and reach contains distinct outcomes. A recommendation can be transmitted without generating adoption; a campaign can enter a new community without reaching many of its members; and early adoption can recruit additional promoters. Comparing only final adoption, or treating all forms of reach as interchangeable, obscures these differences.

A second difficulty concerns the model itself. If tie strength is built to raise cooperation, trust and commitment, then any strategy that prioritizes strong ties is advantaged by construction, and a simulated "strong-tie advantage" partly restates the assumption. The study therefore asks four questions: (1) which strategies generate more adopters and individual coverage, and how does risk change their relative performance; (2) do mixed strategies occupy a favorable conversion–coverage position; (3) does the ranking survive when tie strength no longer determines conversion; and (4) does it survive when total referral effort, rather than effort per promoter, is held fixed? Sensitivity to community mixing, seeding and continuous portfolios is also examined.

A paired simulation design keeps the synthetic population, network, initial promoters and random inputs comparable within each repetition. The experiment is a theoretical exploration, not a prediction for a particular market, consistent with the need to align model assessment with modeling purpose (Edmonds et al., 2019).

1.1. Literature review and analytical expectations

Granovetter (1973) links tie strength and network overlap, explaining why weaker relationships can connect otherwise separated circles. In referral research, Brown and Reingen (1987) distinguish the roles of strong and weak ties across individual and aggregate processes. These arguments motivate a distinction between relationship intensity and intercommunity connectivity; they do not imply that every weak tie is a bridge or that bridge access necessarily produces adoption.

Relational mechanisms also differ across stages. Morgan and Hunt (1994) treat trust and commitment as distinct constructs, and De Bruyn and Lilien (2008) show why word-of-mouth influence should be considered across stages of a recipient's response. Accordingly, the model separates cooperation in transmitting a recommendation, trust in the adoption decision, and commitment in permitting another contact attempt. These are simplified operational assumptions, not empirical estimates or replications. Bampo et al. (2008) distinguish the underlying digital network from the network activated by a campaign; the same distinction applies here, since a structural connection may never carry a recommendation.

Three expectations guide interpretation. First, strong-tie priority may support conversion only insofar as strength covaries with the model's relational attributes. Second, weak-tie/bridge priority may broaden community access, although an advantage in individual coverage is not guaranteed. Third, mixing contact rules may combine local adoption with expansion, but its performance depends on cascade survival and the timing of the mixture. These are mechanism-based questions; the analysis is not a preregistered confirmatory test.

1.2. Perceived risk, network structure and agent-based approaches

Diffusion research has long separated aggregate adoption curves from the micro-level processes that generate them. Aggregate models such as Bass (1969) and the framework summarized by Rogers (2003) describe how adoption spreads through a population but treat the structure of social contacts implicitly. Network-based models make that structure explicit: small-world topologies combine local clustering with short paths (Watts & Strogatz, 1998), and threshold models link individual adoption to the behavior of contacts (Granovetter, 1978). In marketing, agent-based simulation lets researchers combine heterogeneous consumers, explicit contact rules and stochastic processes in settings that analytical models handle poorly, provided that assumptions are documented and that models are verified, validated and examined for sensitivity (Rand & Rust, 2011). The present study follows that guidance by stating every behavioral equation, reporting verification checks and declaring what the model cannot establish.

Empirical and theoretical work on tie strength points in more than one direction. Onnela et al. (2007) report, for a large mobile-communication network, that stronger ties tend to sit within densely connected groups while weaker ties hold groups together; the network used here reproduces that pattern through its mixing parameter and its two tie-strength distributions. Goldenberg et al. (2001) distinguish strong from weak ties in a cellular-automata model of word of mouth and report that weak ties contribute materially to the speed and extent of diffusion despite their lower individual influence. Frenzen and Nakamoto (1993) report that the willingness to pass on market information depends on tie strength and on the stakes attached to the information, which supports treating cooperation as a relational attribute. Centola and Macy (2007) show that when adoption requires reinforcement from several contacts, long ties are less effective carriers than the classic weak-tie argument implies. The present model sits at the opposite end of that contrast: a single recommendation can trigger adoption, and repeated exposure only lowers the probability of adoption. Bridges are therefore not penalized for failing to provide reinforcement, and any advantage of strong ties here cannot come from social reinforcement; it must come from the relational parameters.

Perceived risk adds a further link between relationships and adoption. Taylor (1974) treated perceived risk as a central determinant of consumer behavior, and a common argument in the referral literature is that higher risk raises the value of information from trusted sources. The model encodes this idea through an interaction in which trust offsets the risk penalty. This is a modeling choice, not an empirical estimate, and it determines how risk and tie strength interact in the results; Section 3.4 therefore separates it from the effect of the contact rule itself.

Taken together, these strands suggest three conditional readings of the trust-or-reach choice: strong ties win when relational quality rises with strength; bridges win when access dominates and quality does not depend on strength; and no single rule wins when the effects offset each other. The experiments below are designed to distinguish these readings under identical network, seeding and effort conditions.

2. Methodology

A discrete-time, stochastic agent-based model was implemented in Python. Its purpose is to examine consequences of explicit behavioral and network assumptions. All consumers, relationships, exposures and adoption events are synthetic. The Overview, Design concepts and Details (ODD) summary is in Appendix A, following Grimm et al. (2020).

2.1. Simulation design and data generation

The base design crosses five strategies (A–E) with perceived risk R = 0.20, 0.50 and 0.80. Each of 100 independent repetitions supplies one matched block of all 15 conditions, yielding 1,500 runs. Each run lasts 60 abstract periods (not calibrated calendar months). Each repetition uses 600 consumers in six communities of 100. Undirected edges are sampled independently, with within-community probability $10(1 - \mu)/99$ and between-community probability $10\mu/500$, where $\mu = 0.20$ in the base design. A bridge denotes any intercommunity edge, not necessarily a graph-theoretic cut-edge. A network is accepted only if its active giant component includes at least 570 agents; rejection depends on structure, not diffusion outcomes. The 100 accepted base networks have mean degree 9.93, mean intercommunity edge share 19.58% and mean modularity 0.637.

Twelve initial adopters serve as promoters: six in each of two communities. Within a matched block, all conditions share network topology, consumer and relational attributes, seed identities and indexed random-number inputs. The 1,500 base runs jointly contain 411,030 contact attempts. Repetition r uses seed base $20260925 + 1000(r - 1)$, with separate offsets for network generation, relational attributes, seed selection, dynamic draws and consumer attributes.

2.2. Synthetic consumers, relationships, and strategies

Each consumer j has fixed need N_j and product fit F_j, independently drawn from Uniform(0.5, 1). Tie strength S_{ij} is drawn from Beta(5, 2) within communities and Beta(2, 5) between communities; the distributions overlap. Edges with $S_{ij} < 0.05$ are inactive. Among active edges, the lowest 30% by rank are weak and the highest 30% are strong; the remainder are intermediate. Classes remain fixed throughout a run.

In the base (coupled) condition, trust T_{ij}, cooperation C_{ij} and commitment K_{ij} each satisfy $X_{ij} = 0.70\ S_{ij} + 0.30\ U$ with an independent Uniform(0, 1) draw per attribute. Relational attributes are symmetric, while attempts and histories are directed. No ties are rewired and no attributes are learned.

A satisfied promoter can reserve at most two attempts per period and four over its lifetime. Every reserved attempt consumes one unit of effort, even if no recommendation is transmitted. At the start of a period, eligible recipients are adjacent through active ties, have not yet adopted, and have received fewer than two attempts from that sender. A second attempt can occur only in a later period and becomes eligible with probability K_{ij}; a failed eligibility draw consumes no effort.

Strategy A selects uniformly among eligible strong ties and falls back to all eligible contacts when none are available. Strategy B selects from the first nonempty group in this order: weak intercommunity ties, weak intracommunity ties, other intercommunity ties, all eligible contacts. Strategy C requests route A with probability 0.50 and route B otherwise. Strategy D requests A with probability 0.80 through period 12 and 0.30 thereafter. Strategy E samples uniformly from all eligible contacts. Mixture probabilities apply to each attempted selection and are not exact quotas; requested route and realized tie class are recorded separately.

A reserved attempt by promoter i is transmitted with probability $p(\text{send}_{ij}) = \text{satisfaction}_i(0.20 + 0.80\ C_{ij})$. If the recommendation reaches a consumer who has not adopted, adoption occurs with probability

$$p(\text{adopt}_j \mid \text{sent}) = N_j F_j / [1 + \exp(-z_{ij})],$$

$$z_{ij} = -0.50 + 3T_{ij} - 2R(1 - T_{ij}) - 0.50 \max(0, x_j - 1),$$

where x_j counts prior received recommendations; the redundancy penalty therefore starts with the third exposure. Risk has a nonpositive direct effect by construction, and its interaction with trust raises the trust coefficient to 3 + 2R. After adoption, satisfaction is $0.90(0.50 + 0.50\ F_j)$. Consumers with satisfaction of at least 0.70 receive four effort units and become promoters in the following period; initial promoters have satisfaction 0.90. Resource constraints are therefore equal *per promoter*, while aggregate effort grows endogenously. This base design is not a comparison under a fixed total budget (see Section 2.4).

Within each period, attempts are reserved using the adoption state at its start and processed in randomized order; a recipient who adopts early may still receive another already-reserved recommendation but cannot adopt twice. Newly recruited promoters cannot transmit until the next period. Each adoption retains a parent and root for cascade reconstruction. Extinction means that no promoter retains a viable future opportunity.

2.3. Outcomes and uncertainty

The primary outcome is cumulative new adoption by period 60, excluding the 12 initial adopters. Individual coverage is the number of distinct noninitial consumers exposed divided by 588. Conversion is new adoption divided by all transmitted recommendations. Effort efficiency divides new adoption by realized attempts. Exposure redundancy is repeated exposures divided by transmissions. Community penetration counts previously unseeded communities with at least one new adopter (maximum four in the base seeding). Means give each run equal weight, so mean conversion is the mean of run-level ratios. Contrasts are computed within matched blocks, with 2,000 bootstrap resamples of complete blocks (NumPy PCG64, seed 20261001, 2.5% and 97.5% percentile limits). Intervals are pointwise, not adjusted for multiple comparisons, and quantify Monte Carlo variation under the model, not uncertainty about real consumer behavior.

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A finite conversion–coverage frontier identifies strategies whose mean outcomes are not jointly dominated by another of the five strategies at the same risk level; a nondomination frequency describes resampling stability within this finite set and is not a probability of universal optimality. Verification records for the base experiment cover rule checks, migration checks and reconstruction of outputs from event histories; all 1,500 base runs passed. These checks establish implementation consistency, not empirical validity. No calibration, outcome-based exclusion or fitted coefficient selection was performed.

2.4. Decoupling, effort-budget and sensitivity experiments

Tie-strength decoupling. To separate the effect of the contact rule from the assumption that strength raises relational quality, the three attributes were reassigned across active edges after generation, preserving each attribute's exact marginal distribution and varying only its rank association with tie strength. Four settings were used: *coupled* (original construction); *decoupled* (attributes assigned independently of S); *inverted* (attributes ranked on $0.70(1 - S) + 0.30U$, so weaker ties are more trusting, cooperative and committed); and a graded series in which edges are ranked on $w\,S + (1 - w)U$ for $w = 0.1$, 0.2, 0.3, 0.4 and 0.5, so that $w = 0$ equals decoupled and the original construction ($w = 0.7$ in rank terms) is the upper reference. Strength classes, network, seeds and consumer attributes are identical across settings within a block. Each setting was crossed with all five strategies and three risk levels (45 conditions per block, 100 blocks); the graded series used $R = 0.50$ and strategies A, B, C and E.

Fixed effort budget. A global cap on total referral attempts per run ($B = 100$ or 250, $R = 0.50$) was imposed on top of the per-promoter limits. If the attempts reserved in a period exceed the remaining budget, a random subset is executed; attempts that are not executed consume no promoter effort. Because a cap can fail to bind when promoters exhaust their own effort, realized attempts are reported. Budget 100 binds for all strategies and is the effort-matched comparison.

Sensitivity. Community mixing was varied at $\mu = 0.05$, 0.20, 0.40 and 0.60 ($R = 0.50$; coupled and decoupled; no budget and $B = 250$; 100 blocks each). Seeding was varied across four patterns: six initial promoters in each of two communities (base), twelve in one community, two in each of the six communities, and twelve at random. Finally, a continuous portfolio strategy requested route A with probability q and route B otherwise, for $q = 0, 0.1, \ldots, 1$ ($R = 0.50$, 100 blocks, coupled and decoupled, no budget and $B = 250$). The re-implementation executed 29,500 additional runs; seeds, plans and scripts are archived with the manuscript files. Unlike the base experiment, the re-implementation shares network, attributes and seeds within a block but draws dynamic random numbers independently across conditions.

2.5. Verification of the re-implementation and statistical conventions

The additional experiments were run with a re-implementation of the model written from the rules in Sections 2.1 to 2.4. Following the guidance on rigor in agent-based marketing models (Rand & Rust, 2011), verification took three forms. First, the re-implementation was run under the original specification (coupled attributes, no budget, $\mu = 0.20$, two seeded communities) and compared with the archived results of the original engine (Table A4). Mean new adoption for all five strategies at medium risk lies inside the original 95% bootstrap interval, and coverage, community penetration, attempts and adoption per attempt agree closely. Second, the decoupled and inverted settings preserve the exact marginal distributions of the three relational attributes, so any difference between settings reflects association with tie strength and not the level of trust, cooperation or commitment. Third, the budget rule was checked through realized attempts, which equal the nominal budget whenever the budget binds.

These checks establish consistency with the declared rules and not empirical validity. The re-implementation does not reproduce the indexed random-number scheme of the original engine; it shares network, attributes and seeds within a block, but dynamic draws are independent across conditions. Paired contrasts are therefore noisier than in the original design, and the bootstrap intervals reported for the new experiments are correspondingly wider than they would be under common random numbers. All intervals are pointwise percentile intervals over complete blocks, with no adjustment for multiple comparisons, and the graded coupling series uses a block set independent of the main run.

3. Results

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Results are organized around adoption volume, risk effects, and the distinction between individual coverage and community penetration (Sections 3.1–3.3, original experiment), followed by the new tests of tie-strength coupling, effort, mixing, seeding and portfolios (Sections 3.4–3.7).

3.1. Adoption volume and diffusion trajectories

Table 1 reports final new adoption with 95% bootstrap intervals for its mean (100 runs per cell; the 12 initial adopters are excluded). Mean new adoption follows the same ordering at every risk level: A, D, E, C, B.

Strategy	**Low risk (0.20)**	**Medium risk (0.50)**	**High risk (0.80)**
A: strong priority	149.58 [134.50, 164.47]	115.70 [105.18, 126.41]	89.33 [81.55, 97.18]
B: weak / bridge	23.70 [21.20, 26.35]	18.50 [16.56, 20.58]	14.26 [12.80, 15.89]
C: equal mixture	60.32 [54.32, 66.94]	44.39 [39.87, 49.17]	35.40 [31.70, 39.20]
D: changing mix	89.23 [81.34, 97.70]	70.35 [64.59, 76.69]	57.49 [52.59, 63.10]
E: uniform	69.64 [62.10, 78.27]	52.19 [46.71, 58.16]	38.77 [34.93, 42.78]

Table 1. Mean new adopters by strategy and perceived risk (original engine, coupled attributes), with 95% bootstrap intervals from 2,000 matched-block resamples.

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Under the coupled specification, strong-tie priority generates substantially more diffusion than weak-tie/bridge priority, and the advantage includes the model's permitted fallback contacts. At medium risk, mean attempts are 468.08 for A and 115.49 for B; mean transmitted recommendations are 282.76 and 54.79; and adoption per attempt is 24.04% versus 14.85%. A's larger volume thus combines higher efficiency with a larger endogenously recruited promoter population. Section 3.5 tests performance under equal realized effort.

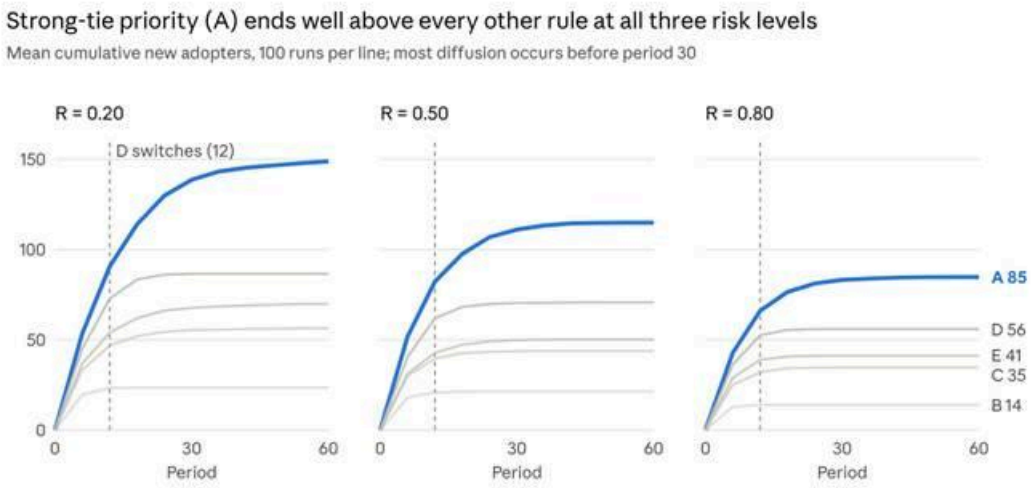


Figure 1. Mean cumulative new adoption over 60 periods. Panels show low, medium and high risk; each curve averages 100 runs and excludes the 12 initial adopters. Curves are descriptive means, not uncertainty bands. The dashed vertical line marks period 12, when strategy D changes its mixing probability. A: strong-tie priority; B: weak-tie/bridge priority; C: equal mixture; D: changing mixture; E: uniform selection. Plotted values come from a fresh replication of the base design and differ from Table 1 by Monte Carlo variation.

3.2. Risk effects and relative strategy performance

Moving from low to high risk reduces mean new adoption by 60.25 for A (95% CI [−71.85, −48.94]), 9.44 for B [−11.41, −7.69], 24.92 for C [−29.44, −20.65], 31.74 for D [−37.06, −26.36] and 30.87 for E [−37.27, −24.95]. All intervals exclude zero, consistent with the negative direct risk term, although not every matched realization decreases because different adoption histories alter later contact opportunities.

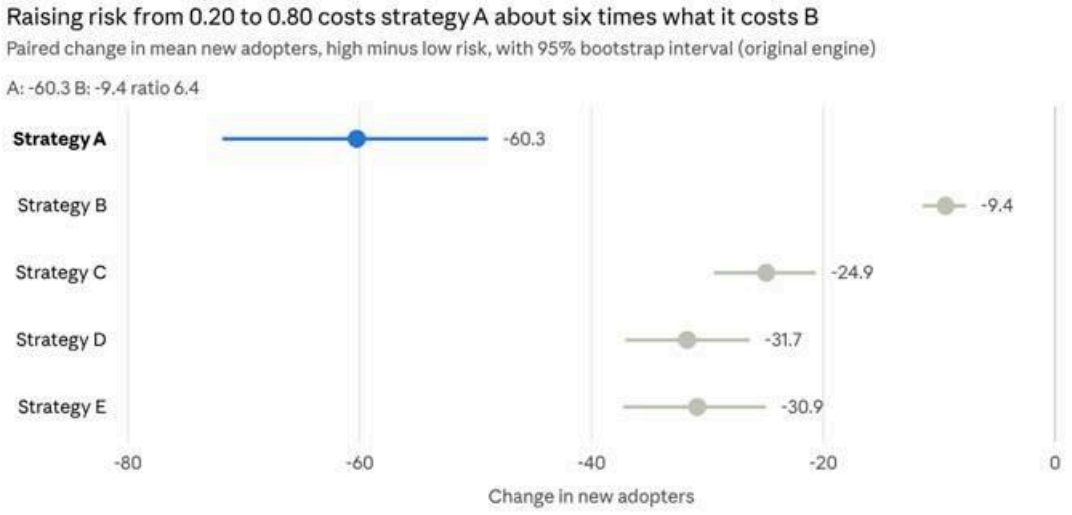


Figure 2. Paired effect of high versus low perceived risk on new adoption. Points are mean within-repetition differences; bars are pointwise 95% percentile bootstrap intervals from 2,000 block resamples. Negative values indicate lower adoption at high risk. All intervals exclude zero.

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Risk changes relative advantages differently across outcomes. A's mean conversion falls from 41.33% to 38.23%, while B's falls from 35.18% to 27.64%; the high-minus-low change in the A−B conversion difference is +4.44 percentage points (95% CI [3.26, 5.66]). In contrast, A's adoption advantage over B declines from 125.88 to 75.07 new consumers (difference in differences −50.81, 95% CI [−62.74, −39.13]). Higher risk widens the conversion gap while narrowing the absolute adoption gap. Because the latent score encodes a smaller risk penalty at higher trust, and trust is coupled to tie strength, this pattern is an implication of the construction; Section 3.4 shows what remains when the coupling is removed.

3.3. Coverage, community penetration, and mixed strategies

At medium risk, individual coverage is 33.32% for A, 23.61% for D, 19.19% for E, 16.82% for C and 8.58% for B. Yet B reaches a mean of 2.84 previously unseeded communities with adoption, compared with 2.49 for A; D and C reach 2.97 and 2.92, and E reaches 2.53. Community penetration and individual coverage therefore need not rank strategies alike. B also has lower exposure redundancy than A (7.65% versus 29.51%), so lower redundancy does not guarantee greater coverage here. Requested routes are not exclusive tie classes: the mean fallback rate is 31.85% for A and 48.89% for B, so the comparison is not one of purely strong versus purely weak contacts.

Early extinction becomes more frequent at high risk. B is extinct by period 12 in 72%, 80% and 92% of low-, medium- and high-risk runs; the proportions for A are 6%, 14% and 15%. D still records attempts after its transition in 89, 87 and 78 runs, so its weaker performance than A is not explained by the second phase never occurring; however, the experiment does not isolate the effect of switching because D already mixes routes before period 13.

At medium and high risk, A alone is nondominated on the conversion–coverage plane. At low risk, A and D are nondominated because D's mean conversion exceeds A's by only 0.08 percentage points (paired interval [−0.67, 0.83]). A is nondominated in all bootstrap mean samples at all three risk levels; D's frequency is 58.95%, 15.40% and 1.00%. This is a finite comparison, not an estimated optimal portfolio.

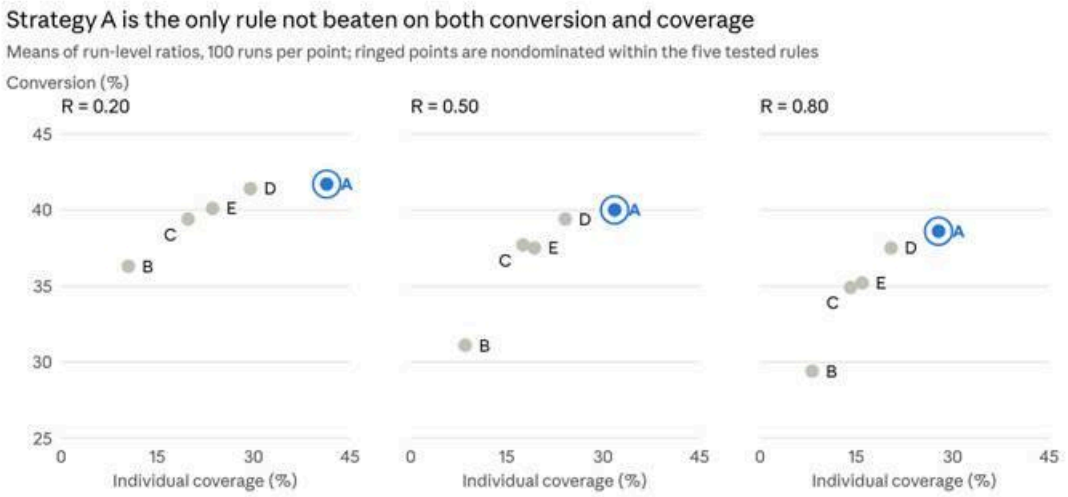


Figure 3. Finite conversion–coverage comparison. Coordinates are means of run-level ratios; labels identify strategies A–E. A ring marks nondominated mean points within each risk panel. In the re-implementation A alone is nondominated at all three risk levels; in the original engine D shares the low-risk frontier because its mean conversion exceeds A's by only 0.08 percentage points, a difference the paired interval does not resolve. Frontier membership does not imply optimality beyond the evaluated strategies.

3.4. Removing the tie-strength tautology: decoupled and inverted attributes

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Table 2 compares the three coupling settings. When trust, cooperation and commitment are independent of tie strength, the large advantage of strong-tie priority disappears. At R = 0.50, A and B produce 45.4 and 44.4 new adopters (A−B = +1.0, 95% CI [−5.5, 7.2]), conversion differs by 0.11 percentage points ([−1.16, 1.31]), and individual coverage is 16.4% versus 17.1%. The intervals for A−B also include zero at low and high risk (+4.4 [−5.6, 14.6] and −0.7 [−5.1, 3.9]). The widening conversion gap under higher risk reported in Section 3.2 does not appear either (A−B conversion +0.57 percentage points at R = 0.80, interval including zero). When the association is inverted, weak-tie/bridge priority dominates: B generates 213.2 new adopters at R = 0.50 versus 22.4 for A (A−B = −190.8 [−205.3, −174.9]).

Setting (R = 0.50, no budget)	A	B	C	D	E	A−B new adopters [95% CI]
Coupled: new adopters	108.6	17.9	46.8	72.0	51.9	+90.7 [79.0, 102.8]
Coupled: coverage (%)	31.8	8.5	17.5	24.1	19.3	
Coupled: unseeded communities reached	2.34	2.71	3.03	2.94	2.51	
Decoupled: new adopters	45.4	44.4	47.1	44.2	40.4	+1.0 [−5.5, 7.2]
Decoupled: coverage (%)	16.4	17.1	18.3	16.9	16.2	
Decoupled: unseeded communities reached	2.06	3.94	3.72	3.25	2.95	
Inverted: new adopters	22.4	213.2	59.5	34.4	46.0	−190.8 [−205.3, −174.9]
Inverted: coverage (%)	9.7	59.0	21.7	13.8	18.1	
Inverted: unseeded communities reached	1.85	3.99	3.89	3.39	3.67	

Table 2. Strategy outcomes under three tie-strength coupling settings (100 blocks per cell; re-implementation).

A−B new adopters across risk levels (low / medium / high): coupled +126.4 / +90.7 / +74.4; decoupled +4.4 / +1.0 / −0.7; inverted −237.8 / −190.8 / −161.3. Under decoupling, B's advantage in community entry becomes large: it reaches 3.94 of four unseeded communities versus 2.06 for A at medium risk. This is the reach advantage that the coupled design conceals, because under coupling bridge access carries a cost in transmission and conversion that offsets it.

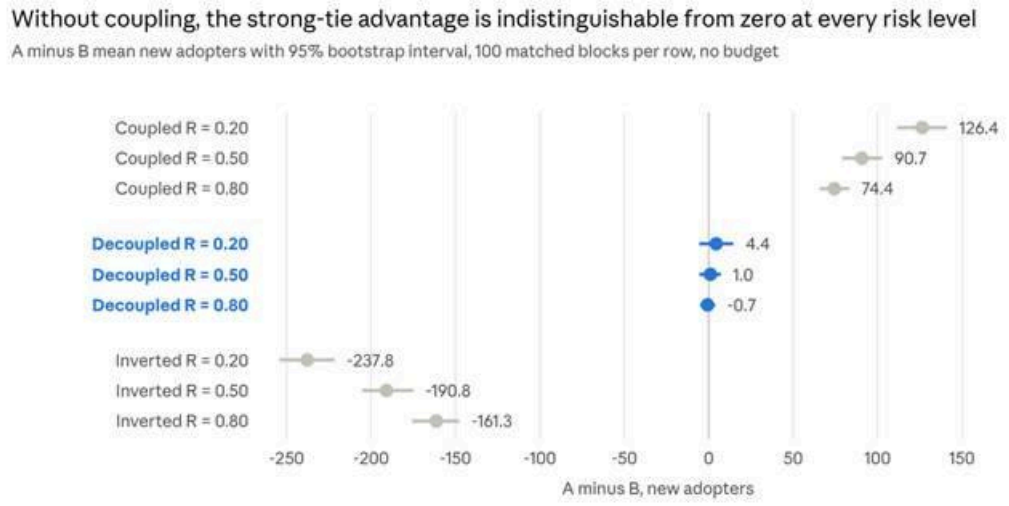


Figure 4. Strong-minus-weak adoption difference (A − B) under three coupling settings and three risk levels. Points are mean paired differences and bars are pointwise 95% percentile bootstrap intervals. Positive values favor strong-tie priority; negative values favor weak-tie/bridge priority.

Rank-coupling weight w	**A−B, no budget [95% CI]**	**A−B, budget 250 [95% CI]**
0 (decoupled)	−1.5 [−7.3, 4.8]	−3.0 [−7.5, 1.3]
0.1	+2.5 [−2.6, 8.1]	+10.9 [6.7, 15.4]
0.2	+15.7 [9.3, 22.2]	+12.8 [8.9, 16.8]
0.3	+33.0 [26.8, 39.9]	+22.0 [18.0, 25.9]
0.4	+52.7 [45.0, 60.4]	+26.4 [22.2, 30.2]
0.5	+64.3 [55.0, 74.1]	+37.1 [33.6, 40.4]
Original coupled	+95.8 [84.9, 106.6]	+42.2 [38.8, 45.4]

Table 3. Strong-minus-weak adoption difference as a function of the strength of association between tie strength and relational attributes (R = 0.50; 100 blocks per row; re-implementation). The graded series uses an independent set of blocks.

The advantage of strong-tie priority increases monotonically with coupling and is first resolved at w of about 0.1 (with a budget) to 0.2 (without). The strong-tie result is therefore a conditional statement about how strongly strength is assumed to determine transmission and conversion, and not a property of strong ties as such.

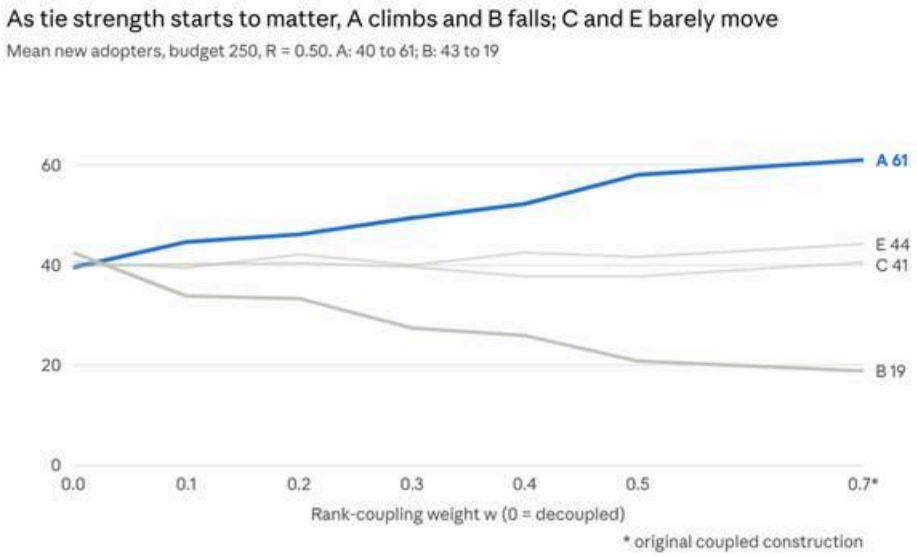


Figure 5. Mean new adopters by strategy as the rank association between tie strength and relational attributes increases (budget 250, R = 0.50). The strong-tie advantage emerges from the gain of A and the loss of B, while the mixed and uniform rules are largely insensitive to coupling. Strategies A, B, C and E were included in the graded series.

3.5. Fixed effort budget

Setting (R = 0.50)	Budget	Realized attempts A / B	New adopters A / B	Adoption per attempt A / B	A−B new adopters [95% CI]
Coupled	100	100 / 96	28.2 / 15.9	28.2% / 16.3%	+12.3 [11.1, 13.6]
Coupled	250	240 / 111	62.4 / 17.2	25.7% / 14.6%	+45.2 [42.3, 47.9]
Coupled	none	441 / 114	108.6 / 17.9	23.9% / 14.8%	+90.7 [79.3, 102.9]
Decoupled	100	100 / 99	21.8 / 21.6	21.9% / 21.6%	+0.2 [−0.9, 1.4]
Decoupled	250	191 / 187	40.2 / 39.7	20.4% / 20.2%	+0.5 [−3.7, 4.8]
Decoupled	none	214 / 210	45.4 / 44.4	20.2% / 20.3%	+1.0 [−5.0, 7.1]

Table 4. Strategy A versus B under a fixed global attempt budget (100 blocks per cell; re-implementation).

With the coupled specification, strategy A remains ahead when realized effort is equal (budget 100: 28.2 versus 15.9 new adopters, and 28.2% versus 16.3% adoption per attempt), so the advantage is not an artifact of A recruiting more promoters. The budget of 250 does not constrain B, which spends only about 111 attempts because its promoters exhaust their own effort; the budget-100 row is therefore the effort-matched comparison. With decoupled attributes, equal effort produces equal adoption: adoption per attempt, which differs by 11 to 12 percentage points under coupling, differs by about 0.2 under decoupling.

3.6. Sensitivity to community mixing and seeding

μ	Coupled, no budget	Coupled, budget 250	Decoupled, no budget	Decoupled, budget 250
0.05	+80.4 [71.1, 89.6]	+35.5 [31.9, 38.8]	−18.2 [−28.2, −7.9]	−2.7 [−6.8, 1.2]
0.20	+90.7 [79.6, 102.1]	+45.2 [42.3, 47.8]	+1.0 [−5.3, 7.2]	+0.5 [−4.0, 4.9]
0.40	+86.5 [76.1, 97.2]	+45.1 [42.2, 47.9]	−0.7 [−4.4, 3.0]	+1.4 [−2.3, 5.3]
0.60	+58.6 [52.8, 64.6]	+42.4 [39.4, 45.2]	−0.5 [−2.9, 1.8]	−1.9 [−4.4, 0.5]

Table 5. A−B new adopters by community mixing (R = 0.50; 100 blocks per cell; re-implementation).

Strong-tie priority leads under coupling at every mixing level, and the strategy ranking A, D, E, C, B is stable for μ up to 0.40 (C and E swap at μ = 0.60). Under decoupling, no difference is resolved except at μ = 0.05 without a budget, where weak-tie/bridge priority outperforms (−18.2 [−28.2, −7.9]). This is the only condition in which a weak-tie advantage in adoption is detected, and it occurs when communities are most separated and strength no longer affects conversion.

Seeding patterns do not change the coupled ranking (A, D, E, C, B in all four patterns). With budget 250, A−B is +37.8 [34.7, 40.5] with twelve seeds in one community, +45.2 [42.3, 47.9] with two communities, +48.7 [46.0, 51.1] with two seeds in each community and +47.6 [44.7, 50.4] with random seeds. Under decoupling the corresponding differences are −4.6 [−9.2, −0.0], +0.5 [−3.9, 4.9], +1.2 [−3.2, 5.4] and −0.4 [−4.9, 4.2], and the order of strategies changes from pattern to pattern with no consistent winner.

3.7. Continuous portfolio of strong and weak priority

Under coupling, mean adoption rises monotonically with the probability q of requesting strong-tie priority: from 18.3 (q = 0) through 46.0 (q = 0.5) to 121.0 (q = 1) without a budget, and from 19.0 through 41.0 to 64.6 with a budget of 250. Pure strong-tie priority has the highest mean in 99% and 100% of bootstrap resamples, respectively. Efficiency and individual coverage rise in the same way. No intermediate mixture therefore outperforms pure strong-tie priority, and the finite result for strategy D (Section 3.3) is consistent with a continuous search. Under decoupling the portfolio curve is essentially flat (43.4 at q = 0, 42.9 at q = 0.5 and 41.6 at q = 1 without a budget; 43.6, 42.0 and 37.3 with a budget of 250), and the maximizing q is unstable across resamples (q = 0.6 in 55% of resamples without a budget; q = 0 in 39% with a budget). Weak-tie/bridge priority continues to enter more unseeded communities as q falls (3.85 at q = 0 versus 1.87 at q = 1 without a budget).

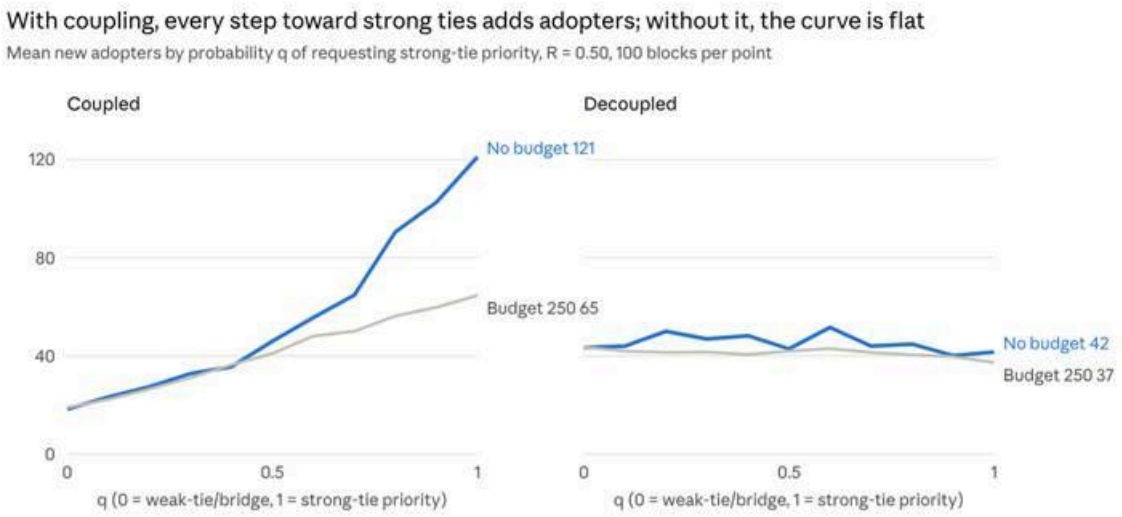


Figure 6. Mean new adoption along a continuous portfolio of strong-tie and weak-tie/bridge priority. Under coupling the maximum is at q = 1 in 99% (no budget) and 100% (budget 250) of bootstrap resamples; under decoupling the maximizing q is unstable across resamples.

4. Discussion

The main finding is conditional. When tie strength covaries with transmission, trust and commitment, prioritizing strong ties sustains more adoption and greater individual coverage than prioritizing weak ties and intercommunity contacts, and it does so under every risk level, mixing level, seeding pattern and, for the coupled specification, under equal realized effort. The decoupling experiments show, however, that this ordering is carried by the assumed relationship between strength and conversion. With that relationship removed, the two rules perform alike in adoption, conversion and coverage, and the weak-tie/bridge rule enters more previously unseeded communities; with the relationship reversed, the ranking reverses. The model does not discover that strong ties work better; it shows how large an advantage emerges once the cascade dynamics process an assumed relational advantage.

This result does not eliminate the structural role of weak ties described by Granovetter (1973). It separates access to communities from diffusion intensity. Under coupling, B enters about as many or more unseeded communities than A while reaching much fewer individuals. Under decoupling the access advantage becomes large (3.94 versus 2.06 of four communities at medium risk) without a cost in volume, which suggests that the original design neutralized the structural benefit of bridges by tying it to lower trust and commitment. The graded series gives a practical reading: in this model, an association between strength and relational quality as weak as a rank weight of 0.1 to 0.2 is enough for the strong-tie advantage to be resolved. Whether real referral networks exhibit such an association, and with what magnitude, is an empirical question that the simulation cannot answer.

The mechanism that survives is cumulative recruitment under finite effort. Under coupling and equal realized effort (budget 100), strong-tie priority still converts more attempts into adopters (28.2% versus 16.3%), so the advantage is not explained solely by A recruiting a larger promoter population. Under decoupling the corresponding efficiencies are equal. The effort-matched comparison also clarifies a limitation of fixed budgets in this model: a budget that exceeds what a rule's promoters can spend (B at 250) does not bind, so equal nominal budgets do not guarantee equal realized effort.

The risk comparison adds a qualification. The latent adoption score assigns a smaller risk penalty when trust is greater, and trust is coupled to strength. The widening conversion gap and narrowing adoption gap reported in Section 3.2 are therefore implications of this encoded interaction, and they vanish under decoupling. What requires simulation is the resulting cascade behavior across heterogeneous consumers, overlapping contact opportunities and finite promoter lifetimes. A relative advantage can increase on one scale while decreasing on another, so entrepreneurial interpretations should specify whether the objective is adoption volume, conversion efficiency or entry into new groups.

The mixed strategies do not provide clear evidence of a superior balance. Strategy D ranks second in adoption at every risk level, and the continuous portfolio search shows that, under coupling, adoption rises monotonically with the weight on strong-tie priority; under decoupling the portfolio curve is flat and its maximum is unstable. A mixture does not automatically capture the best of both routes. The timing of the switch in D remains untested, because D already mixes routes before period 13.

Relation to prior work. The results qualify, rather than contradict, the weak-tie argument. Granovetter (1973) concerns the structural position of weak ties. The present experiments show that this structural advantage is visible in community entry when relational quality is independent of strength (3.94 versus 2.06 unseeded communities at medium risk), and that it is overwhelmed when weaker ties are also less cooperative, less trusting and less committed. Empirical patterns such as those described by Onnela et al. (2007), in which weak ties connect groups, are compatible with both outcomes; which one applies depends on the relationship between strength and the propensity to refer, a relationship that Frenzen and Nakamoto (1993) suggest may itself depend on the stakes of the information. Centola and Macy (2007) show that reinforcement requirements can reduce the value of long ties. The present model contains no reinforcement mechanism, so it is a conservative test of the strong-tie advantage: any gain for strong ties here comes from relational quality and not from reinforcement. Models that add reinforcement would plausibly shift the balance further toward strong ties, but this was not tested.

The seeding experiments connect to the influence-maximization literature, in which the choice of initial adopters is treated as an optimization problem (Kempe et al., 2003). The present experiments vary only a handful of fixed seeding patterns with random promoters within them, so they show robustness of the ranking and not an optimal seeding policy.

Implications for entrepreneurial marketing. The findings do not support a general rule that ventures should favor close contacts, nor one that they should seek outsiders. They support a diagnostic: before allocating referral effort, estimate whether the contacts reachable through weaker or bridging ties are less willing to pass on and adopt than close contacts, and by how much. If the gap is large, concentrating effort on close contacts raises adoption per unit of effort, as in the coupled budget-100 condition. If the gap is small, effort spent on bridges buys access to new communities at little cost in conversion, which matters when the objective is market entry rather than volume. Because the model ignores the cost of reaching distant contacts, it is conservative on this point. A venture that values both volume and entry should also note that mixing the two rules did not produce a better compromise than the pure rules: in the coupled case the mixtures lay between the pure rules, and in the decoupled case they were indistinguishable from them. Finally, higher perceived risk shrinks the absolute gains of every rule, while the relative advantage of close contacts can grow even as its absolute size falls, so the objective function determines the lesson.

Several limitations define the scope of the evidence. The decoupling is performed by rank reassignment that preserves marginal distributions; other functional forms, such as diminishing returns to strength or interactions between strength and bridging, are not explored. Alternative seeding patterns and community mixing change one factor at a time, and results do not establish robustness to network size, degree or network family. Tie strength in the original design covaries with several favorable attributes at once, so their individual contributions are not separated. Effort per promoter, product quality, the satisfaction threshold and the redundancy penalty are fixed; neither learning nor network evolution is included. The new experiments used an independent re-implementation and a design in which dynamic random draws are not shared across conditions within a block, so they should be re-run with the original engine. The parameters are transparent academic assumptions, without empirical calibration or claims of market realism, and increasing the number of runs reduces simulation sampling uncertainty but not uncertainty about those assumptions. Future work should vary trust weight, effort, redundancy and the switching time of mixed rules, and should confront the coupling parameter with empirical evidence on how tie strength relates to recommendation transmission and adoption.

5. Conclusion

In 1,500 base runs arranged as 100 matched blocks, strong-tie priority generates the highest mean adoption and individual coverage across three risk levels, while weak-tie/bridge priority achieves somewhat broader community entry at medium risk without greater population diffusion. Higher risk reduces adoption under every strategy, widens the strong-tie conversion advantage and narrows its absolute adoption advantage. Additional experiments show that this ranking depends on the assumed association between tie strength and relational quality: it disappears when the association is removed (with weak-tie priority reaching more unseeded communities) and reverses when it is inverted. Under coupling, the advantage persists at equal realized effort, across community mixing and seeding patterns, and across a continuous portfolio of strong and weak priority, where no intermediate mixture outperforms pure strong-tie priority.

The study's contribution is a reproducible separation of adoption volume, individual coverage, community penetration and conversion, and of the assumptions that connect them. Its implications remain conditional on the synthetic network, the behavioral equations and the per-promoter effort constraints. Further variation in network structure and behavioral assumptions, and empirical evidence on the strength–conversion relationship, are needed before advancing a broader theoretical claim or a practical prescription.

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Contribution statement

Author names and CRediT roles remain to be assigned and approved by the authors. The contribution statement will distinguish conceptualization, methodology, software, verification, analysis, visualization, writing, and supervision as applicable.

Statement of data consent

No human participants or personal records were used; all data were generated by simulation. Consent to collect participant data is therefore not applicable to this study. This statement does not represent an institutional ethics approval.

The version 0.2 engine, experimental plans, data manifests, audit records, and analysis outputs are maintained in the project files. The risk-experiment index links the extension and its baseline provenance: https://drive.google.com/file/d/16SO5Ox70XGNUOsQLXn0yZE8dLSuLQAHU/view. Access depends on the project's sharing permissions. A permanent public repository and data/code licensing should be agreed by the authors before submission; public availability is not claimed in this draft.

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Appendices

Appendix A. ODD summary and reproducibility details

A1. Purpose and patterns

The model explores how contact selection, perceived risk and the coupling between tie strength and relational attributes influence synthetic referral cascades. Patterns of interest are adoption trajectories, individual coverage, conversion, entry into unseeded communities and extinction; they are comparison outputs, not empirical calibration targets.

A2. Entities, state variables, and scales

Consumers carry community identity, need, fit, adoption status and period, satisfaction, remaining effort, exposure history and cascade ancestry. Undirected relationships carry strength, trust, cooperation, commitment, active status, bridge status and strength class. Directed sender–recipient histories record attempt counts and timing. The system contains 600 consumers, six communities and 60 periods; there is no geographic metric or calibrated time scale.

A3. Process overview and scheduling

Initialize network, attributes and 12 seeds. At each period start, determine promoter eligibility and recipients from a snapshot of adoption. Apply commitment to repeat-contact eligibility, select routes and contacts, and reserve effort; when a global budget is active, trim reserved attempts at random to the remaining budget. Randomize reserved-attempt processing, determine transmission and adoption, and recruit satisfied adopters for the next period. Record states through period 60.

A4. Design concepts

Diffusion and aggregate effort emerge from local stochastic decisions and promoter recruitment. Consumers do not learn or optimize. Strategies use local eligibility, fixed strength class and community membership, with no global novelty oracle. Interactions occur through existing active edges.

A5. Initialization

Generate independent within- and between-community edges at the probabilities in Section 2.1. Apply the activity threshold and giant-component acceptance criterion. Draw consumer need and fit independently. Rank active ties to define weak and strong classes. Generate baseline relational attributes, then derive coupled, graded, decoupled and inverted settings by rank reassignment of the same attribute values. Select seeds according to the seeding pattern, initialize satisfaction to 0.90 and effort to four, and retain the same initialization across the conditions of a repetition.

A6. Input data

There are no exogenous empirical inputs or time-varying external series. Risk is a fixed treatment for a run. Quality, contact capacity, effort, relational dependence and the coefficients in Section 2.2 are prescribed academic assumptions. The independent sample consists of repetitions (blocks), not individual consumers or event records.

A7. Submodels and reproducibility

The contact hierarchy, transmission equation, logistic adoption equation, satisfaction equation, delayed promoter activation and extinction rule are specified in Section 2.2. For the base experiment, seed offsets use candidate indices 0–99 for topology and 100–199 for relationship attributes, followed by 200 for seed selection, 201 for dynamic inputs and 202 for consumer attributes, with dynamic draws indexed by period, sender and contact slot. The original environment is Python 3.12; the engine and event auditor use the standard library, and analysis uses archived NumPy and Matplotlib dependencies. The base experiment is defined by run_risk.py and analyze_risk.py and their manifests. The additional experiments use abm.py, experiments.py, sweep.py and analysis.py (bootstrap seed 20261001 for the main analysis). Validation of the re-implementation against the base medium-risk results (A / B / C / D / E new adopters: 108.6 / 17.9 / 46.8 / 72.0 / 51.9 versus 115.70 / 18.50 / 44.39 / 70.35 / 52.19) is a consistency check, not a proof of equivalence.

A8. Parameters, strategies, outcomes and replication check

Tables A1 to A3 summarize the model specification of Section 2; Table A4 reports the verification described in Section 2.5.

Component	Specification	Role in the model
Population and time	600 consumers; 6 communities of 100; 60 periods	Fixed scale of every run
Network	Expected degree 10; within-community edge probability $10(1 - \mu)/99$; between-community probability $10\mu/500$; $\mu = 0.20$ in the base design	Community structure and bridge density
Tie strength	Beta(5, 2) within communities; Beta(2, 5) between; edges below 0.05 inactive	Overlapping strength distributions by tie type
Strength classes	Lowest 30% of active ties weak; highest 30% strong	Defines routes A and B
Relational attributes	T, C, K = 0.70 S + 0.30 U (coupled); rank-reassigned for graded, decoupled and inverted settings	Trust, cooperation, commitment
Consumers	Need and fit independent Uniform(0.5, 1)	Adoption probability and satisfaction
Seeds	12 initial adopters: six in each of two communities (base)	Initial promoters, satisfaction 0.90
Effort	4 units per promoter; at most 2 attempts per period; at most 2 attempts per sender–recipient pair	Per-promoter resource constraint
Transmission	satisfaction × (0.20 + 0.80 C)	Probability that a reserved attempt becomes a recommendation

Volume, issue, pages and DOI: pending publisher assignment.

Adoption	N F / [1 + exp(−z)], z = −0.50 + 3T − 2R(1 − T) − 0.50 max(0, x − 1)	Trust, risk and redundancy
Recruitment	Satisfaction 0.90 (0.50 + 0.50 F); promoter if at least 0.70	Endogenous growth in campaign effort
Risk levels	R = 0.20, 0.50, 0.80	Treatment fixed within a run
Replication	100 matched blocks; 2,000 bootstrap resamples	Monte Carlo uncertainty

Table A1. Model parameters and their role.

Strategy	Contact rule	Fallback
A: strong priority	Uniform among eligible strong ties	All eligible contacts
B: weak / bridge	Weak intercommunity ties, then weak intracommunity, then other intercommunity	All eligible contacts
C: equal mixture	Route A with probability 0.50, otherwise route B	As in the chosen route
D: changing mix	Route A with probability 0.80 through period 12, then 0.30	As in the chosen route
E: uniform	Uniform among all eligible contacts	None required
Portfolio (new)	Route A with probability q, otherwise route B, q = 0, 0.1, …, 1	As in the chosen route

Table A2. Contact-selection strategies.

Outcome	Definition
New adoption	Cumulative adopters by period 60, excluding the 12 initial adopters
Individual coverage	Distinct noninitial consumers exposed, divided by 588
Conversion	New adoption divided by all transmitted recommendations
Effort efficiency	New adoption divided by realized attempts
Exposure redundancy	Repeated exposures divided by transmissions
Community penetration	Previously unseeded communities with at least one new adopter
Extinction	No promoter retains a viable future opportunity

Table A3. Outcome measures.

Medium risk (R = 0.50)	Strategy	Original engine	Re-implementation
New adopters	A	115.70 [105.18, 126.41]	108.6

New adopters	B	18.50 [16.56, 20.58]	17.9
New adopters	C	44.39 [39.87, 49.17]	46.8
New adopters	D	70.35 [64.59, 76.69]	72.0
New adopters	E	52.19 [46.71, 58.16]	51.9
Individual coverage (%)	A / B	33.32 / 8.58	31.8 / 8.5
Individual coverage (%)	C / D / E	16.82 / 23.61 / 19.19	17.5 / 24.1 / 19.3
Unseeded communities reached	A / B	2.49 / 2.84	2.34 / 2.71
Unseeded communities reached	C / D / E	2.92 / 2.97 / 2.53	3.03 / 2.94 / 2.51
Mean attempts	A / B	468.08 / 115.49	441 / 114
Adoption per attempt (%)	A / B	24.04 / 14.85	23.9 / 14.8

Table A4. Replication check of the re-implementation against the original engine (100 blocks each; original intervals in brackets).

Appendix B. Remaining scope before a submission version

Author identities, affiliations, correspondence details, funding, conflicts, CRediT roles, and final data-sharing terms require author confirmation. The reference list covers the foundations cited here; any relationship to earlier manuscripts or a doctoral thesis should be documented using the actual source before submission.